

1 Status of the Hall A Møller Polarimeter

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1.1 Introduction

The Hall A Møller polarimeter [1] had been built in 1997. It was successfully used to measure a beam polarization for all Hall A experiments with polarized electron beam.

1.2 General description

The Møller scattering events are detected with a magnetic spectrometer (see Fig.1) consisting of a sequence of three quadrupole magnets and a dipole magnet. The electrons scattered in a plane close to the horizontal plane are transported by the quadrupole magnets to the entrance of the dipole which deflects the electrons down, toward the detector. The optics of the spectrometer is optimized in order to maximize the acceptance for pairs scattered at about 90° in CM. The acceptance depends on the beam energy. The typical range for the accepted polar and azimuthal angles in CM is $75^\circ < \theta_{CM} < 105^\circ$ and $-6^\circ < \phi_{CM} < 6^\circ$. The non-scattered electron beam passes through a 4 cm diameter hole in a vertical steel plate 6 cm thick, which is positioned at the central plane of the dipole and provides a magnetic shielding for the beam area. The plate, combined with the magnet's poles, make two 4 cm wide gaps, which serve as two θ_{CM} angle collimators for the scattered electrons. Two additional lead collimators restrict the ϕ_{CM} angle range. The polarimeter can be used at beam energies from 0.8 to 6 GeV, by setting the appropriate fields in the magnets. The lower limit is defined by a drop of the acceptance at lower energies, while the upper limit depends mainly on the magnetic shielding of the beam area inside the dipole.

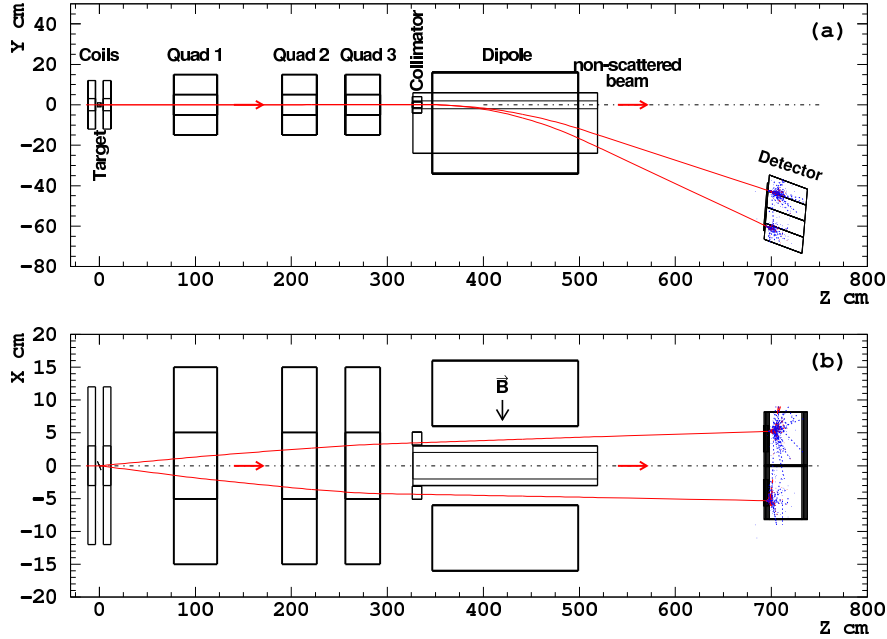


Figure 1: Layout of the Møller polarimeter before 11 GeV upgrade, (a) presents the side view while (b) presents the top view. The trajectories displayed belong to a simulated event of Møller scattering at $\theta_{CM} = 80^\circ$ and $\phi_{CM} = 0^\circ$, at a beam energy of 4 GeV.

The detector consists of total absorption calorimeter modules, split into two arms in order to detect two scattered electrons in coincidence. There are two aperture plastic scintillator detectors at the face of the

calorimeter. The beam helicity driven asymmetry of the coincidence counting rate (typically about 10^5 Hz) is used to derive the beam polarization. Additionally to detecting the counting rates, about 300 Hz of “minimum bias” events containing the amplitudes and timings of all the signals involved are recorded with a soft trigger from one of the arms. These data are used for various checks and tuning, and also for studying of the non-Møller background. The estimated background level of the coincidence rate is below 1 %.

1.3 12 GeV Upgrade Status

The Hall A Møller polarimeter originally was designed for an electron beam energy range of 1 – 6 GeV. Two factors limit the useful energy range of the polarimeter:

- the spectrometer acceptance, defined by the positions of the magnets and the available field strength, and also the positions and of the collimators;
- the beam deflection in the Møller dipole caused by the residual field in the shielding insertion.

In order to operate the polarimeter at 11 GeV a considerable upgrade of the polarimeter was required. In order to minimize the interference of such an upgrade with the rest of the beam line we did not consider moving the Møller target or the Møller dipole magnet and the Møller detector, as well as replacing the shielding insertion in the dipole magnet.

A few items have to be considered for the higher energy polarimeter design:

1. the positions and settings of the quadrupole magnets;
2. the dipole magnet bending angle;
3. the dipole shielding insertion;
4. the detector position;
5. the beam line downstream of the Møller dipole.

1.3.1 Quadrupole magnets position

The acceptance of a Møller polarimeter is defined as the accepted range of the scattering angles in CM, around 90° . In Hall A polarimeter a collimator, consisting of two vertical slits between the poles of the dipole magnet and the shielding insertion in the dipole gap plays the most important role in limiting the acceptance. The goal of the quadrupole magnets is to direct the scattered electrons into the slits. With the old (6 GeV) design, two quadrupole magnets (*PATSY* and *FELICIA* see Table 1) were used.

GEANT simulation shows that for 11 GeV era power of two and even all three existing Møller quadrupole magnets is not enough. In order to cover the new beam energy range of 0.8 – 11 GeV we proposed to move the first quadrupole 40 cm downstream and to install the fourth quadrupole with its center at 70 cm from the Møller target.

The new quadrupole magnet was designed by Robin Wines. The magnet is shown on Fig. 2. The new quadrupole has been field mapped by Ken Bagget [2] before installation on the Hall A beam line and the results for the new magnet are presented in Table 1. A new bench was designed and manufactured to install the new quadrupole and to shift the first magnet (*PATSY*). A distance between the Møller target and the new quadrupole magnet center is 0.684 m. A distance between the Møller target and the first Møller quadrupole magnet is 1.334 m. The second and the third Møller quadrupole magnets position is unchanged.

Available *Danfysik* power supply from Accelerator Division will be used to power the new Møller quadrupole magnet to save money. It is already installed and working, but the EPICS controls have not been done. It is a work in progress.

Table 1: Parameters of the Møller quadrupole magnets.

Møller notation	Q0	Q1	Q2	Q3
MCC notation	<i>MQO1H01</i>	<i>MQM1H02</i>	<i>MQO1H03</i>	<i>MQO1H03A</i>
Name	<i>new</i>	<i>PATSY</i>	<i>TESSA</i>	<i>FELICIA</i>
Bore, cm	10.16	10.16	10.16	10.16
Effective length, cm	36.58	44.76	35.66	35.66
Maximum current, A	300	300	280	280
Pole tip field at 300 A, kGs	6.39	5.94	6.03	6.14

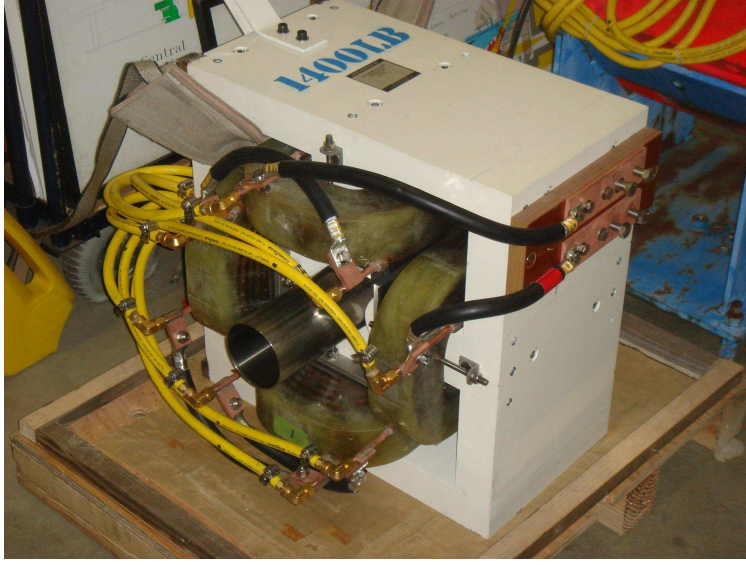


Figure 2: New quadrupole magnet for Møller polarimeter.

1.3.2 Dipole bending angle

The old Møller electrons bending angle in the dipole is 10° . A dipole current of about 700 A and a field of about 19.2 kGs is needed to keep this bending angle at 11 GeV. The maximal magnetic field measured in this dipole in Los Alamos was 17.5 kGs. The present dipole power supply provides the maximal current of 550 A. This current is not enough to provide for the beam bending angle in dipole of 10° at the beam energy 11 GeV. This limitation, along with the problem of shielding the beam area at high fields, described below, is mitigated by reducing the bending angle from 10° to 7.0° . The smaller bending angle allows to keep the existing Møller dipole and its power supply. The reduction of the bending angle requires a new detector position, as it will be described below.

1.3.3 Dipole shielding insertion design

The dipole shielding insertion attenuates the strong dipole magnetic field in the region where the main electron beam passes through the dipole. It was designed for the dipole magnetic field up to 10 kGs. This field is enough to bend the Møller electrons to the Møller detector at a beam energy of 6 GeV. For a higher beam energy and a stronger magnetic field the shielding insertion becomes saturated leading to a strong residual field and a large deflection of the electron beam..

The diameter of the bore in the shielding insertion is 4.0 cm. The diameter of the electron beam line before and after the Møller polarimeter is 2.54 cm. A coaxial magnetically isolated pipe, made of magnetic steel AISI-1010, was placed inside the bore (see Fig. 3) to increase the attenuation of the shielding insertion.

The inner pipe diameter is 2.5 cm and the outer diameter is 3.4 cm. The shielding pipe consists of eight assembled together sections to reduce the cost. The shielding pipe is centered in the shielding insertion bore with seven isolating rings made of a non-magnetic aluminum 6061-T6. The total shielding pipe length is 202.4 cm. It is about 15 cm longer than the shielding insertion length in order to reduce the influence of the fringe field outside of the shielding insertion.



Figure 3: Møller dipole assembly with additional shielding pipe in the shielding insertion.

The new design allows to attenuate to an acceptable level the dipole magnetic field up to 14.8 kGs. A field of 14.0 kGs (and power supply current 513 A) corresponds to the beam energy of 11 GeV and the dipole bending angle 7.0° . This field can be provided with the existing power supply.

The TOSCA simulated fields in the dipole gap, in the shielding pipe and the expected electron beam shift on the Hall A target and in the beam dump are shown in Fig. 4. A new vertical corrector is installed downstream of the Møller dipole (see Sec. 1.3.5) to compensate the beam shift at high beam energies.

1.3.4 Detector position and shielding

Because of the smaller bending angle of the Møller electrons the detector has to be lifted by 10 cm. The beam line downstream of the dipole also has to be modified. Originally, it was planned to re-use the old detector shielding box with some modifications. It occurred that the design of the old box was in conflict with the design of a new beam line girder downstream of the Møller detector. A new shielding box was designed, manufactured and installed at the new position on the Hall A beam line (see Fig. 5).

Before the upgrade the beam pipe diameter after the Møller dipole was 6.35 cm (2.5 inches). The beam pipe diameter over the detector shielding box was 10.16 cm (4 inches), and after that (girder area) 2.54 cm (1 inch). After the upgrade one 6.35 cm (2.5 inches) pipe is used between the Møller detector and the beam line girder.

Lead bricks on the top of the shielding box and along the beam line downstream of the Møller dipole have been reassembled in accordance with the new beam line design.

1.3.5 New girder design downstream of the Møller dipole

Precise knowledge of the beam position and angle on the Møller target is important for the optimal beam tuning and for understanding of the systematic errors of the beam polarization measurements. The old beam

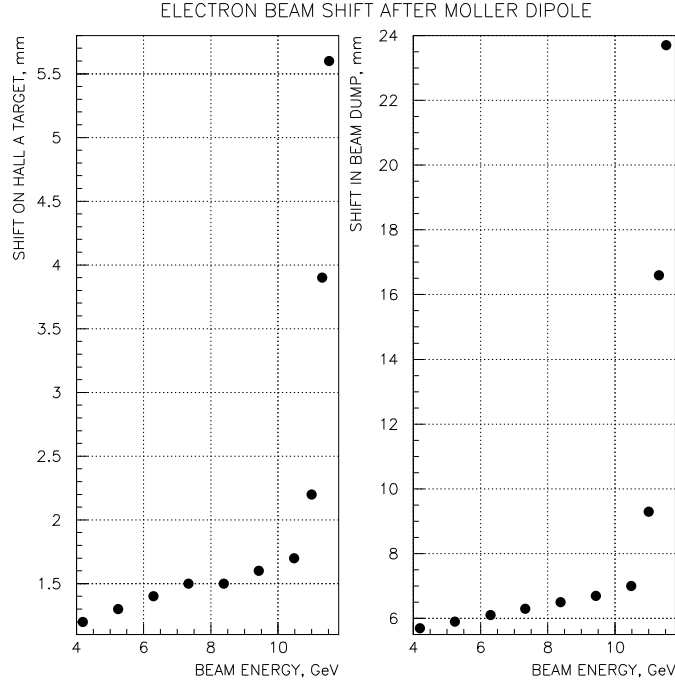


Figure 4: TOSCA result for the Møller dipole with the 10 cm extended shielding pipe. The electron beam shift on the Hall A target (left picture) and in the Hall A beam dump (right picture).

line provided only three BPMs for the position/angle measurements:

- BPM *IPM1H01* - in 1 m upstream of the Møller target;
- BPM *IPM1H04A* - upstream of the Hall A target (in 17 m downstream of the Møller target);
- BPM *IPM1H04B* - in the Hall A beam dump.

There were three (at least two) Møller quadrupole magnets, Møller dipole, two quadrupole magnets downstream of the Møller detector and a few beam position correctors between BPM *IPM1H01* and BPM *IPM1H04A*. Because of that precise information about the beam position and especially beam angle on the Møller target and good beam tuning was not available.

In the new beam line design a new BPM (see Fig. 6) is installed on the girder downstream of the Møller detector. The new BPM is located ~ 7 m downstream of the Møller target. Centering of the beam with the Møller quadrupole magnets and dipole should provide correct beam tuning for the beam polarization measurement and precise information about the beam position and angle on the Møller target.

At high energies the shielding insertion in the Møller dipole is saturated and the residual field deflects the beam down (see Fig. 4). A new vertical corrector (see Fig. 6) was installed on the girder to compensate for this effect.

1.4 Møller polarized electron targets

Magnetized ferromagnetic materials are used to provide polarized electrons in the target. The average electron polarization in such targets is about 7-8%. It is not theoretically calculable with an accuracy sufficient for polarimetry, and has to be somehow measured. The uncertainty of this value is typically the dominant systematic error of the the Møller polarimetry. Two different techniques to magnetize ferromagnetic targets

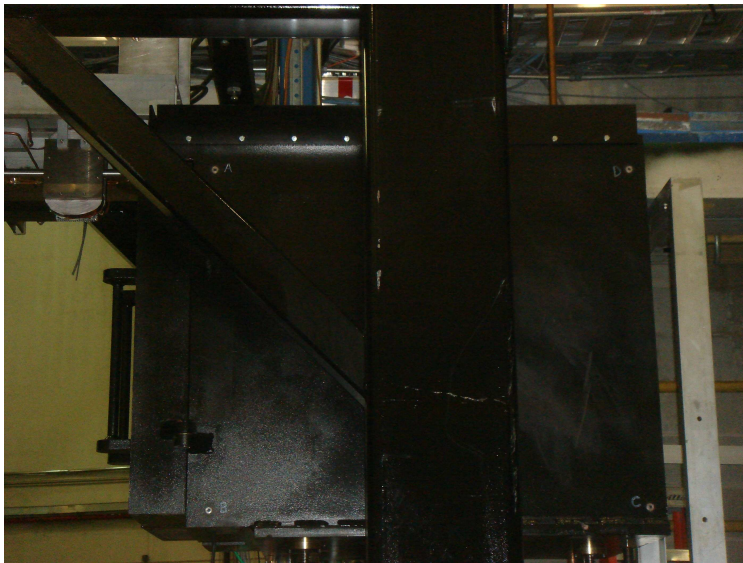


Figure 5: A new Møller detector shielding box on the Hall A beam line.

are used.. The first one - the “low field” technique - uses a thin ferromagnetic foil tilted at a small angle to the beam and magnetized in the foil’s plane by a relatively weak magnetic field (~ 20 mT) directed along the beam. The second one - the “high field” technique - uses a thin ferromagnetic foil positioned perpendicular to the beam and polarized perpendicular to its plane by a very strong field (~ 3 T). Description and comparison of both types of the polarized electron targets can be found in [3]. The Hall A Møller polarimeter is a unique polarimeter which uses both this techniques. This allows a better understanding of the systematic error associated with target polarization.

1.4.1 “Low field” polarized electron target status

A detailed description of the “low field” target is done in [4]. The target was used with the Hall A Møller polarimeter in 2005 - 2009. The target consists of six foils, of Supermendure and iron with different thickness from $6.8 \mu\text{m}$ to $29.4 \mu\text{m}$, fixed at an angle of 20.5° to the beam in the YZ plane, magnetized by a $B_Z \sim 0.035$ T field.

The target holder design is shown on Fig. 7. The holder can move the targets across the beam in two projections: transversely - along X , and longitudinally - along the longer sides of the foils (a line in the YZ -plane, at 20.5° to Z). The goal is to study the observed effects of non-uniformity of the target magnetic flux, measured by a small pickup coil at different locations along the foil. Systematic error budget for the Møller polarimeter with the “low field” polarized electron target is presented in Tab. 2

In the beginning of 2011 after PREX and DVCS experiments the “low field” target was restored back to the Møller polarimeter for the beam polarization measurements for g2p experiment. There were a few reasons to choose the “low field” target for g2p experiment:

- g2p experiment does not require high precision of the beam polarization measurement;
- g2p experiment was running with very low beam current 0.1 mA. A maximal efficient thickness of the “high field” target is $\sim 10 \mu\text{m}$. A maximal efficient thickness of the “low field” target is $\sim 90 \mu\text{m}$. Thus, using of “low field” target allows to reduce essentially time required for the beam polarization measurement with the same statistical error;
- operation of the “low field” target is cheaper because it does not require expensive cryogenics;
- operation of the “high field” target at present requires a daily accesses to the Hall to feed the target superconducting magnet.

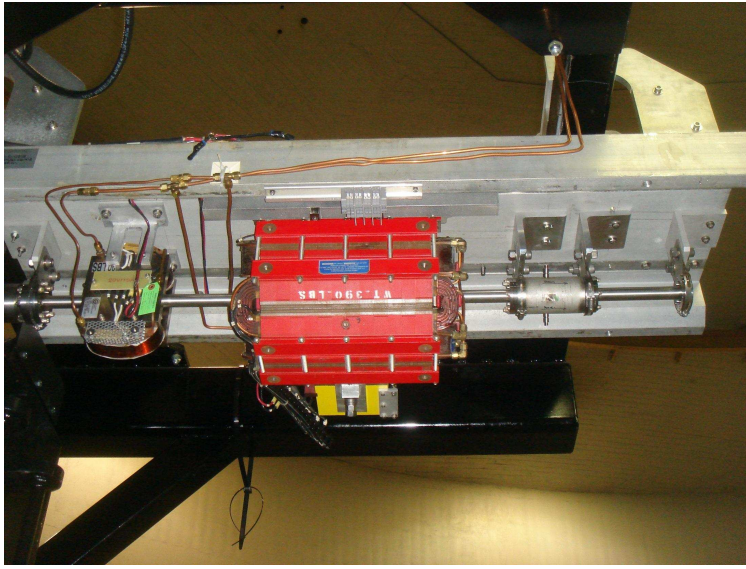


Figure 6: A new girder downstream of the Møller detector shielding box on the Hall A beam line. From left to right: new vertical corrector *MBD1H04*, focusing quadrupole magnet *MQAH04* and new beam position monitor *IPM1H04*.

The "low field" target was successfully used during the running of the g2p experiment. The "low field" target is installed on the Hall A beam line now and it will be used for the Møller polarimeter commissioning after the 11 GeV upgrade.

1.4.2 "High field" polarized electron target status

Experiment PREX required a polarimeter accuracy of $\sim 1\%$. As it is shown in Tab. 2, the Møller polarimeter with the "low field" target can not meet the requirement. Instead, a new "high field" polarized electron target for the Hall A Møller was built. The "high field" technique [5] uses a strong magnetic field - larger than the magnetic field inside of the ferromagnetic domains. The field should orient the magnetization in the domains along the field direction and drive the magnetization into saturation. In the polarimeter, the magnetic field is parallel to the beam direction. The foil is perpendicular to the field, in order to minimize the effects of the magnetization in the foil plane, and is magnetized perpendicular to its plane. The value of the magnetization (and of the average electron polarization) at saturation depends only on the material properties, and for pure iron can be derived from the existing world data [6].

Table 2: Systematic errors for the Hall A Møller polarimeter with the "low field" and the "high field" polarized electron targets.

Variable	"Low field"	"High field"
Target polarization	1.5%	0.35%
Analyzing power	0.3%	0.3%
Levchuk-effect	0.2%	0.3%
Background	0.3%	0.3%
Dead time	0.3%	0.3%
High beam current	0.2%	0.2%
Others	0.5%	0.5%
Total	1.7%	0.9%

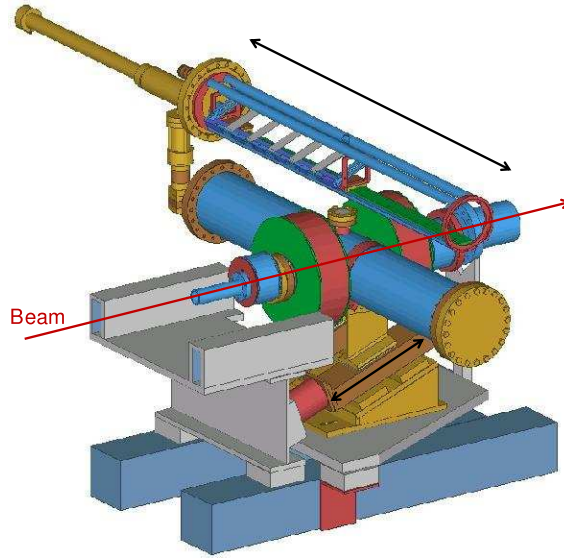


Figure 7: The "low field" target holder design. The electron beam direction and directions of the target motion in two projections are shown.

Design of the "high field" polarized electron target is shown on Fig. 8. The target consists of:

- a superconducting magnet for a maximal magnetic field of 4 T. The magnet needs liquid He^4 at low pressure;
- a target holder with a set of four iron foils with the purity of 99.85% and 99.99%. The foils thicknesses are 1,4,4 and 10 μm to study possible sources of systematic errors (see Fig. 8);
- a mechanism of target foils orientation along the magnetized field;
- a mechanism for targets motion into the beam;
- a mechanism of the magnetic field orientation along the beam.

The "high field" target was used in 2010 for the beam polarization measurements during the PREX and DVCS experiments running. As it is seen from Tab. 2 using of the "high field" target allows to increase the accuracy of the beam polarization measurements by a factor of two. It should be noted that a successful operation of the "high field" requires considerable efforts:

- improvements of the target foils and magnetized field alignment;
- gaining the target operation experience;
- a systematic error study;
- building a supply line for liquid He^4 .

1.5 Møller polarimeter DAQs

The Hall A Møller polarimeter has two DAQs:

- old DAQ based on combination of CAMAC and VME modules;
- new DAQ based on FADC.

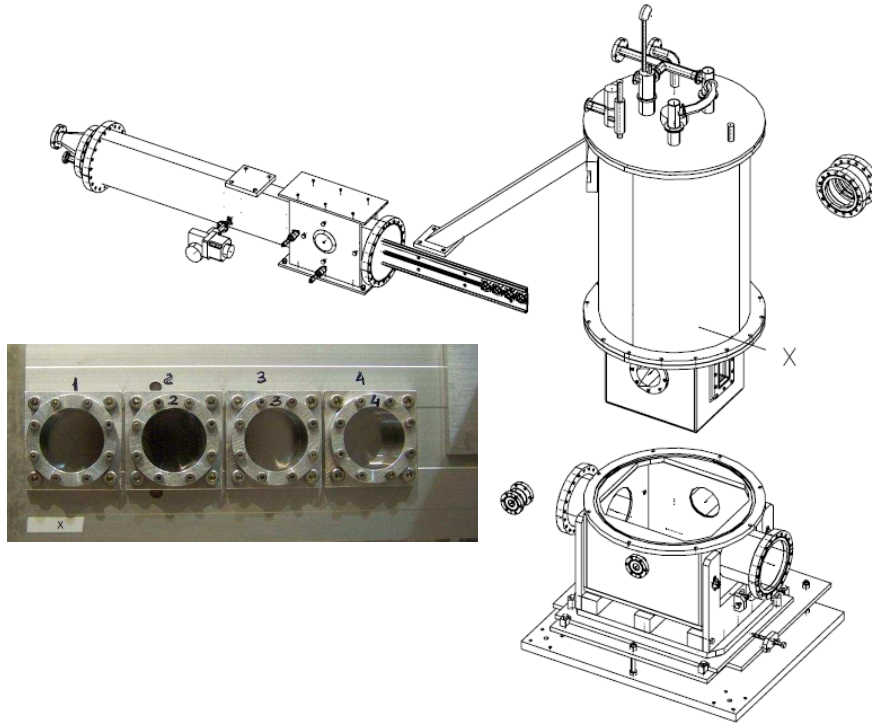


Figure 8: Design of the "high field" polarized electron target. The target holder with four pure iron foils is shown on photo.

The old DAQ is fully operational with both polarized electron targets, well understood but slow, occupies a few crates and uses a few hundred cables to connect modules etc. New DAQ based on FADC is fast, generates two types of triggers, compact, but not fully operational yet. Running of two different DAQs in parallel and comparison of the results gives a unique opportunity to study possible sources of systematic errors.

1.5.1 Old Møller DAQ upgrade status

Present DAQ for the Moller polarimeter detector has been designed in the mid-90th. It uses a lot of slow modules not available in stock anymore. The main goals of the electronics upgrade for the Moller polarimeter are:

- to increase bandwidth (up to 200 MHz) of the detector system;
- to reduce readout time from ADC and TDC modules;
- to replace the old PLU module LeCroy-2365 that is not available in stock anymore.

The list of modules to be replaced:

- to increase bandwidth:
 - PLU module LeCroy-2365, bandwidth < 75 MHz, CAMAC replaced with PLU module based on CAEN V1495 board (bandwidth 200 MHz, VME);
 - Discriminator Ortec-TD8000, input rate < 150 MHz, CAMAC replaced with P/S 706 (300 MHz, NIM), modified for remote threshold setup with DAC type of VMIC4140;

- to reduce readout time:
 - ADC LeCroy 2249A, 12 channels, CAMAC replaced with QDC CAEN V792 (32 channels, VME);
 - TDC LeCroy 2229, CAMAC replaced with TDC V1190B (64 channels, 0.1 ns, VME).

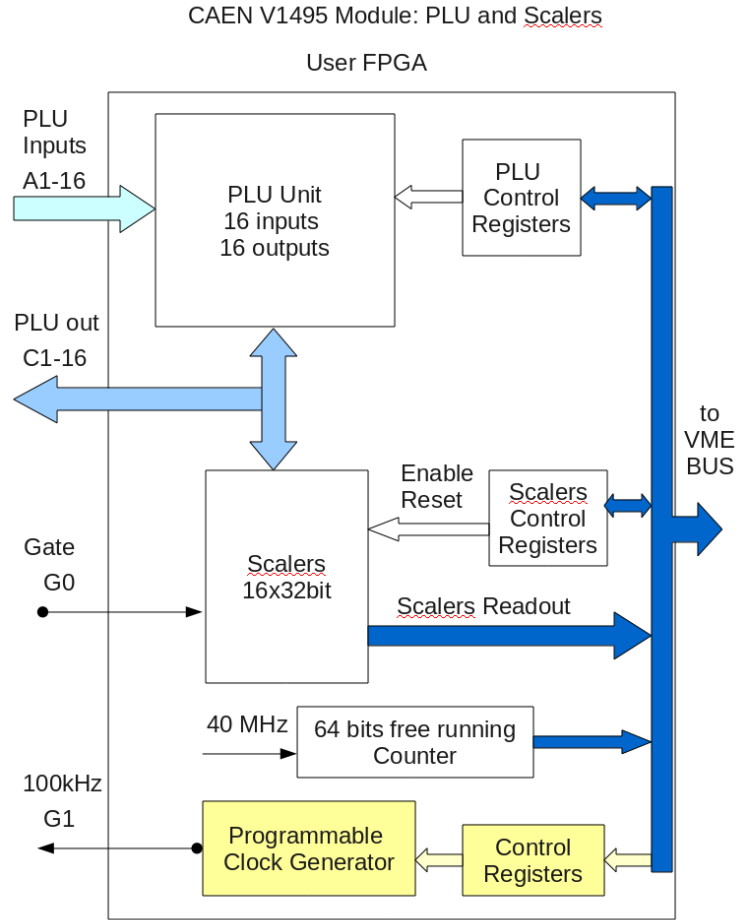


Figure 9: PLU diagram for CAEN V1495 module.

Diagram for new PLU unit based on CAEN V1495 module is shown on Fig. 9. The module CAEN V1495 has the following parameters:

- Input bandwidth ~ 200 MHz;
- 2 input ports x32 bits;
- 1 output port x32 bits;
- 2 input/output front LEMO connectors;
- The FPGA "User" can be reprogrammed by the user using custom logic functions.

Firmware for the PLU module is under development and will consists of the following units:

- Programmable Logical Unit (PLU): 16 inputs, 16 outputs;

- Scalers unit: 16 channels, 32 bit, gate input, connected to PLU outputs;
- Free running 64 bit timer with base frequency 40 MHz.

All the modules required for the upgrade have been procured. The work is in progress. We plan to use the old DAQ after the upgrade at least until the new DAQ based on flash-ADC will be fully operational with both the low and high field targets (see details below in Sec. 1.5.2). Also, running of two different DAQs in parallel provides a unique opportunity to study systematic errors.

1.5.2 Status of FADC DAQ for the Moller detector

A new DAQ based on the JLab-built FADC was created in 2009 for PREX experiment to be operated with the new high field polarized electron target (see [7], [8]). The schematics of the new DAQ is shown on Fig. 10.

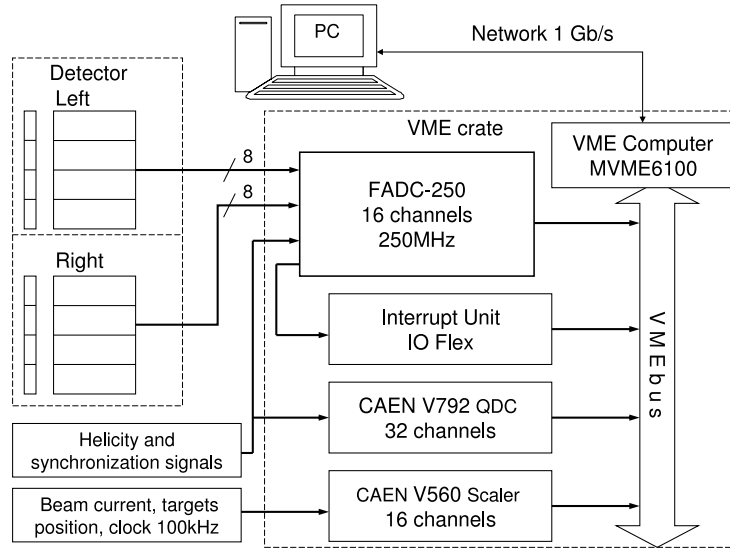


Figure 10: Scheme of a new Møller DAQ based on FADC.

There are some differences between the old and the new DAQ due to differences between the low field and the high field targets operation. For the low field target, the target polarization is a function of the particular foil, the foil coordinate and the magnetic field of the magnetized Helmholtz coils. The direction of the magnetic field is flipped every run to reduce the systematic error. For each run the old DAQ with analyzer is doing the following:

- ramps up the current in the Helmholtz coils;
- reads out the value of the current;
- starts the data taking when the field is established;
- reads out of the foil number and the coordinate of the foil on the beam line;
- reads out of the beam position;
- turns off the current in the Helmholtz coils when the required number of events has been acquired;
- calculates the foil polarization for the particular place of the foil and for the particular magnetizing field and the field direction;

- uses the calculated foil polarization for the beam polarization calculation.

There are two versions of analyzers to run the old DAQ with the low field and the high field targets.

As it was mentioned above, FADC was built to run with the high field target. For this configuration the target is fully saturated and the target polarization is a constant for any foil, foil coordinate and magnetic field. Magnetizing field in the superconducting magnet is turned on in the beginning of the beam polarization measurement and turned off in the end of the measurements. The FADC DAQ does not perform some functions needed for the low field running and, therefore, for the low field running the old DAQ is mandatory while the new DAQ is optional.

The DAQ based on the FADC generates two types of triggers:

1. helicity flipping triggers (integral mode / scalers);
2. data triggers (single events).

There is a good agreement between the old and the new DAQs in scalers mode (see Fig. 11.)

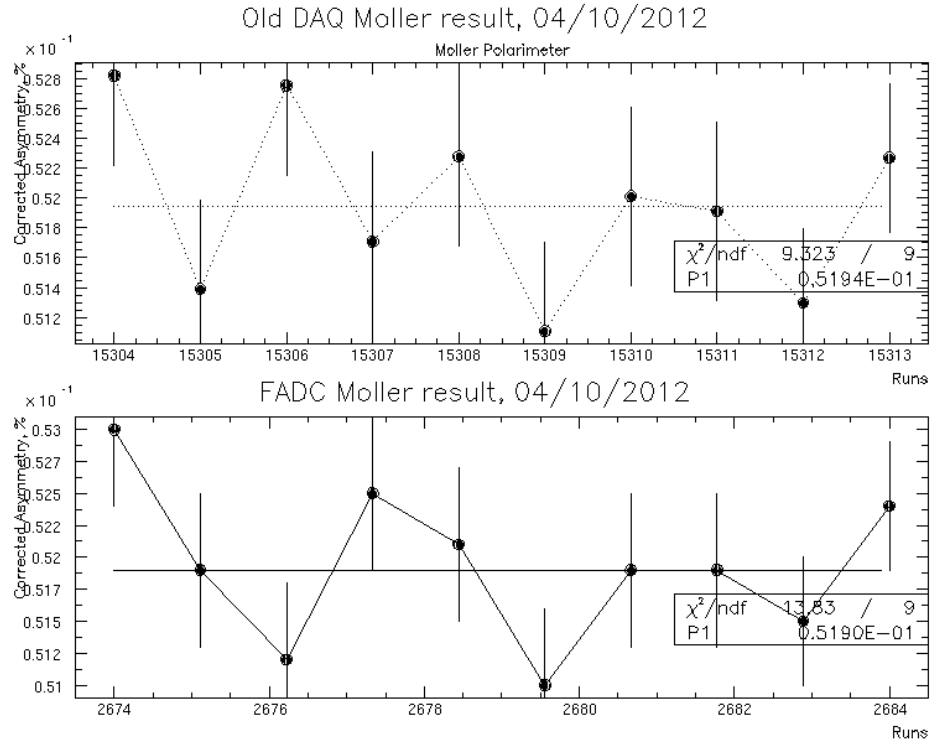


Figure 11: Comparison of results of the beam polarization measurements with the old and the new Møller DAQs in the scaler mode.

Running of the FADC in the data trigger mode is important to study the systematic errors. The triggers data should help to:

- improve the GEANT model of the polarimeter;
- increase the accuracy of the evaluation of the average analyzing power;
- study the Levchuk-effect.

At the moment, the work on the event data analysis is in progress.

1.6 Summary

The beamline part of the Møller polarimeter 11 GeV upgrade is completed. The polarimeter can be operated in the beam energy range of 0.8 - 11.0 GeV. The Møller polarimeter is ready for commissioning with the beam. The remaining work includes modifications and checkout of the DAQ system, the "high field" target, a cryogenics line to feed the "high field" magnet, and the documentation for the Møller polarimeter operations after upgrade.

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